EXECUTION-PATH QUALIFICATION AND REALIZED COSTS IN SPECULATIVE DECODING ON CONSUMER SYSTEMS

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ABSTRACT

Numerical execution paths can change greedy output, but observing a mismatch does not predict the rest of a trajectory. On the tested B570 stack, we select a draft-free target-only micro-batch-one predictor using exposed discovery data, then freeze complete generated-ID predictions before reserved live speculation. It matches 72/72 capped sequences on 24 repeat-stable prompts, including all 12 runs of four reference-divergent prompts; the same-set reference matches 60/72. The predictor is sufficient for this path, but it neither restores qualification nor identifies a unique numerical cause. Separate B570 same-state forks test intervention effects; RTX proposal/replay programs test reproduction while sharing numerical routines. Cost comparisons answer a separate question. Newly direct-ID-qualified RTX serialization is slower within its instrumented post-prefill boundary (0.646× target/serialized latency). At fixed measured progress and target work, 358H remains below parity even with draft and residual costs removed. Qualified M4 Guard beats target-only but loses to fixed speculation. These results give testable diagnoses and engineering decisions about lossless compatibility and complete cost on the measured paths; they establish no portable decoding or control gain.

1 INTRODUCTION

Speculative decoding uses cheap proposals to amortize target work (Leviathan et al., 2023). Prior consumer-hardware measurements show both speedups and slowdowns, as well as greedy argmax changes when the target's batch shape changes (Chordiya, 2026). We ask whether an execution-path hypothesis can predict complete, unseen speculative output, including departures from the intended reference.

We test this on one B570 Vulkan stack. A draft-free target-only UB1 predictor is selected after 36/36 discovery matches; its 72 complete ID arrays are then frozen before reserved live outcomes. They match 72/72 runs on 24 repeat-stable prompts, including four reference-divergent prompts (12 runs); the UB128 reference matches 60/72. A wrong later token would falsify the prediction. The result shows that target-only execution suffices to predict this path, without identifying a unique kernel cause. It also gives a clear decision for lossless optimization: this path cannot replace the reference, even when its output can be predicted accurately.

Two supporting studies examine causality and transfer. A separately selected B570 cohort compares identical fork states: scores change but immediate decisions do not, while some later trajectories differ under a persistent alternate policy. RTX's target-only H1 fails discovery. A later, independently orchestrated P/R program reproduces reserved live paths using sealed proposals and shared numerical routines. It uses drafting to construct predictions and does not show that proposal/history inputs are necessary. These studies address questions that the original draft-free prediction alone cannot answer.

A compatible path still needs a separate cost test. Original direct-ID gates qualify RTX serialization but leave B570 blocked; they do not certify historical timing. A new RTX study qualifies its own timer path, yet serialization is slower. On 358H, measured target work prevents arithmetic parity at fixed progress even with draft cost eliminated. On M4, hindsight selection between fixed modes leaves little opportunity, and Guard adds no aggregate benefit. Each case uses its own frozen comparator, so these results do not rank the hardware.

Our three contributions are (1) direct-ID path qualification and matched instrumented cost on a newly qualified RTX path, where compatibility does not bring a speedup; (2) prospective B570 full-trajectory prediction, with separate intervention and RTX program tests of its causal and transfer limits; and (3) complete 358H interface accounting and qualified M4 fixed-policy comparisons that show the limits of draft-only and continuation-control improvements.

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2 Experimental Question and Measurement Discipline

Acceptance, isolated draft throughput and target-forward batching measure different boundaries. None alone measures the complete speculative loop. We distinguish them using an output contract and complete-cost accounting.

2.1 A path-qualified output contract

Let k be the maximum draft proposal width, UB the target micro-batch capacity (n_{ubatch}), and EOG a vocabulary end-of-generation ID. A greedy speculative round verifies draft proposals, commits the matching prefix and a correction or bonus when budget permits, then removes inconsistent suffix state. Our strict contract requires committed generated IDs to equal the matched target-only greedy reference under the frozen configuration. We can therefore test lossless compatibility with a specified deployed greedy reference for each model, runtime, path and workload. A failed pair blocks a workload-global optimization claim; a small passing probe only establishes eligibility for broader testing. Capped ID equality does not measure functional HumanEval correctness or pass@k.

Historical text-path agreement (H) compares arrays reconstructed from parsed CLI text. Direct-ID qualification (Q) compares captured IDs against the study's frozen reference. Frozen prediction equality (P) compares a sealed prediction with reserved live IDs. A predictor can match a reference-divergent path, so P does not imply Q. Same-state intervention separately compares scores, immediate decisions and subsequent trajectories; these endpoints must not be interchanged.

2.2 Evidence classes

Primary results answer a frozen question after its applicable gate; **secondary** results retain their registered or post-study follow-up scope. **Mechanism-supported** results use path controls or measured regions. **Diagnostic** values describe blocked or restricted paths. Historical RTX controls remain text-path secondary evidence and B570 timing diagnostic, irrespective of later direct-ID results.

2.3 Complete round costs

Let T_b denote matched target-only cost per output token, T_d^{real} the complete realized draft-subsystem time including interface replay, $T_{\text{target,round}}^{\text{real}}$ complete target work, and T_x residual synchronization, transfer and bookkeeping outside those regions. Count all committed output, including correction/bonus, as C. Then

$$S = \frac{T_b C}{T_d^{\text{real}} + T_{\text{target,round}}^{\text{real}} + T_x} \tag{1}$$

$$= \frac{C}{D_{\text{real}} + R_{\text{target}} + X}. \tag{2}$$

Terms in the second denominator are normalized by T_b; break-even requires $C > D_{\text{real}} + R_{\text{target}} + X$. These are accounting identities, not new profitability theory (Kim et al., 2026). Aggregate means and totals require consistent weights; total-time ratios need not equal prompt geometric means. Accepted drafts alone do not count correction/bonus output. Incompatible experiments cannot isolate an unmeasured component by subtraction.

2.4 A separate target-forward proxy

For context n and path e,

$$V_k^{\text{micro}}(n, e) = \frac{T_{\text{target forward}}(k, n, e)}{T_{\text{target decode}}(1, n)}. \tag{3}$$

This narrower microbenchmark omits drafting and may omit correction/bonus and state work; V_k^{micro} is not complete R_{target}. Isolated draft throughput likewise cannot substitute for stateful D_{real}. These measurements help diagnose the tested path but do not identify a universal cause.

3 Experimental Methodology and Platforms

3.1 Frozen systems and workloads

Each study was frozen independently; we compare them retrospectively. Table 1 lists questions, comparators and sample units. Appendix K gives historical hardware settings and follow-up configurations. RTX's original direct-ID OS build is 26200; the new H1, H2 and latency studies use 26300. Driver versions alone do not identify driver bytes. Final comparisons process one request at a time. The target/draft pair is Qwen3-8B/0.6B Q8_0, except the 358H adapter's symmetric INT4 OpenVINO draft.

The representative M4 study selects 24 unseen Spec-Bench prompts across six groups (Xia et al., 2024). A separate preregistration selects 30 HumanEval tasks before input inspection (Chen et al., 2021); fixed $k = 2$ is primary and threshold variants secondary. Three repetitions and a 128-ID cap apply to the code studies. The prospective runtime studies retain their separate task splits and estimator in Section 7.

For direct-ID studies, "complete" means all committed/generated IDs through the first inclusive EOG or 128-ID cap, excluding prompt IDs and uncommitted proposals. Original B570 H1 reserved arrays all reach the cap; RTX H2

Table 1. Study map. Counts are distinct prompts × repetitions; complete ID endpoints include the first EOG or cap128. H = historical text-path comparison, Q = reference qualification, P = frozen prediction. Timing is matched only within the named study. M4 uses Metal, RTX CUDA, B570 Vulkan; 358H combines Vulkan target and OpenVINO draft. Details and evidence entries are in the separate appendix.

Question / study	Reference or contrast	Prompts × reps	Endpoint / role	Matched timing
Historical costs	M4, RTX, B570 own baselines	24/30 × 3	Parsed text (H)	Historical only
Direct-ID gates	RTX UB1; B570 UB128	30 × 3 each	All IDs (Q)	None
B570 H1 prediction	UB1 predictor; UB128 reference	24 × 3	72/72 P; Q blocked 60/72	None
B570 live fork	Same state UB1 vs UB128	New 24 × 3	Scores, decisions, later IDs	None
RTX H1 discovery	Target-only UB2 vs live	12 × 3	24/36; reserved 0	None
RTX V2 P/R replay	Sealed proposals; UB1 reference	New 24 × 3	72/72 P; Q blocked 60/72	None
RTX serialized cost	Own target-only UB1	30 × 3	90/90 Q; latency	Instrumented CLI
358H Phase 3	Own target-only; CPU/NPU draft	30 × 3	90/90 Q per path; cost	Complete loop
M4 runtime control	Target, Always; fixed-mode oracle	60 × 3	180/180 Q per policy	Charged loop

includes genuine earlier EOGs. Each follow-up uses its own reference and selected tasks. Exposure-informed discovery is distinct from reserved inference: audit and deterministic selection precede reserved execution, and predictions/tapes are sealed before reserved live. No-known-exposure means no record in the audited scope, not proof about unavailable host files. The appendix preserves selection, freezes, denominators and endpoint distributions.

Timing excludes loading and initial prefill. Native historical counters and the 358H complete-loop timer keep their original boundaries. The new RTX timer starts after synchronized target/draft prefill and before per-request sampler and generation-state work, ending at slot release; its instrumented CLI boundary is detailed in Section 5.4. Different runtimes, quantization and references preclude hardware-only causal comparisons.

3.2 Statistical estimators and uncertainty

Historical M4/RTX/B570 compute the GM of three paired ratios for each prompt; 358H uses their median. Headline speedups are prompt GMs with 95% prompt-bootstrap intervals. M4 HumanEval uses 100,000 resamples, the other hardware campaigns 10,000. New RTX latency instead takes each condition's three-time median, then target/serialized ratio and prompt GM. Its 10,000-draw prompt bootstrap preserves frozen seed, order and linear percentiles. M4 runtime has its own ordered streams. Repetitions and within-prompt decision events are correlated; they are not independent prompts. Intervals condition on the campaign and do not include full session, thermal or recalibration uncertainty.

3.3 Provenance and energy boundaries

An evidence ledger maps each study's canonical bytes and freeze roles. Hashes identify content; server acknowledgements and host records supply reported ordering, not trusted external process time. Anonymous derivatives preserve arrays but cannot authenticate original absolute chronology or omitted weights/runtime/compiler/driver bytes. Appendix O lists exact packages and capabilities; offline checks execute no models.

M4 energy measurements cover the package aggregate and exclude separate DRAM; B570 measurements cover the adapter board. RTX mixed domains are not a common system boundary. No end-to-end 358H Phase-3 or runtime energy result is claimed. Historical Windows per-run provenance is weaker; 358H source reconstruction is strongly supported by retained hashes and code behavior, but exact historical identity is unproven (the internal `STRONGLY_SUPPORTED` label). No platform energy ranking follows.

Existing-record analysis. We calculate the score distributions, fixed-mode selection opportunity and conditional cost bounds after the study, with aggregate outcomes already known. A separate plan fixes inputs, ordering and a new sensitivity seed before recomputation; no original endpoint or confidence interval is replaced. Appendix P gives the formulas and identifies the anonymous inputs.

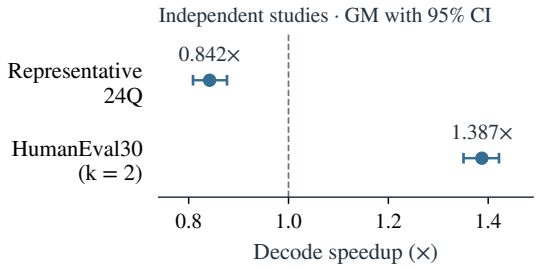


Figure 1. **Opposite profitability regimes on the same tested M4 stack**. Representative 24Q yields 0.842× [0.809, 0.877], while the separately preregistered HumanEval30 fixed-$k = 2$ study yields 1.387× [1.351, 1.422]. Points are campaign prompt geometric means of paired decode speedups; intervals are campaign-specific 95% prompt-bootstrap intervals. The dashed line marks parity. Both pass the recorded re-tokenized-text comparator (72/72 and 90/90 pairs); these are not direct generated-ID comparisons or functional HumanEval pass@k. These are separate studies, not a paired between-workload comparison; differing workloads and session conditions do not isolate workload-only causality.

4 Gains Do Not Translate Uniformly

Table 2 reports historical timing under recorded text-path agreement for M4/RTX/B570. Fresh direct-ID qualification appears separately in Section 5.1 and Figure 3; it does not convert those timings into strict-ID performance results. The final RTX block reports fresh qualification and matched instrumented latency (Section 5.4); it is a different campaign. A blocked path remains diagnostic.

4.1 M4: outcomes in two separate studies

The representative frozen M4 sample produces 0.842× speedup (95% CI [0.809, 0.877]), with 72/72 text-path-equal pairs and only 1/24 prompts faster. The separately preregistered HumanEval fixed-$k = 2$ study produces 1.387× [1.351, 1.422], with 90/90 text-path-equal pairs and 30/30 prompts faster. Figure 1 retains both; the positive code-conditioned result does not replace or pool with the negative representative result.

The two studies observe opposite outcomes on the same hardware, with different workloads and sessions. These are not workload-only causal effects or general M4 profitability. The discovery effect size is not a confirmatory estimate; early MLX failures do not negate the historical Q8_0 Metal result.

For primary fixed-$k = 2$ HumanEval, the package-efficiency ratio (baseline/speculative J per output token) is 1.013× [0.990, 1.034]. The interval includes parity; a package-energy improvement is not established. The registered $p_{\min} = 0.9$ condition has secondary energy evidence, discussed separately in the appendix. The preregistered corrected synthetic ceiling was not executed. We retain that deviation rather than treating the earlier synthetic diagnostic as its replacement.

4.2 Across stacks: negative and blocked outcomes

On RTX 5070 Ti Laptop, the serialized CPU-to-RTX control reaches 0.404× [0.390, 0.418], while the same-GPU draft control reaches 0.650× [0.629, 0.673]. Both pass 90/90 text-path comparisons and accelerate zero of 30 prompts. The higher speed of same-GPU placement shows that placement and associated integration costs matter, but its continued loss rules out transfer as the sole explanation.

On Arc B570, neither the fast path nor the serialized fallback supports a workload-global admissible optimization result. The latter's diagnostic speedup is 0.3087× [0.2927, 0.3246], with zero of 30 prompts faster. That interval describes timing on a partially mismatching path; its precision does not turn it into an admissible speed claim. The formal failure and stop rule remain the platform result. Board-energy observations likewise remain diagnostic and boundary-limited.

On Core Ultra X7 358H, the closed Phase-3 heterogeneous runner has direct-ID sequence equality on both CPU and NPU drafting: 90/90 pairs per path, with zero of 30 prompts faster. CPU achieves 0.734× [0.693, 0.771]; NPU achieves 0.352× [0.333, 0.369]. Both qualified paths are slower despite favorable isolated draft measurements. Section 5 examines the measured costs behind this slowdown.

5 From Local Opportunity to Realized Gain

We test output qualification, frozen path prediction and sampled intervention effects, then examine realized cost on a qualified path. Figure 2 shows the original B570 discovery, freeze and reserved test; later studies use their own tasks and gates.

5.1 Execution-path qualification

Figure 3A retains eleven historical target-forward proxy cells. Metal $V_2^{\text{micro}} \approx 1.02$–$1.08$, near-step RTX batching, and B570 fast 1.24–1.64 versus serialized 1.983–1.992 describe different runs and contexts. Lower proxy values mean cheaper target-forward work relative to one-token decode. These measurements omit integration, cannot replace complete R_{target}, and have no same-study matched cost/qualification tradeoff with Panel B.

The original direct-ID studies each complete 270 executions and 180 comparisons over 30 HumanEval prompts × three, requiring 90/90 reference equality. RTX matches 75/90 batched and 90/90 serialized; only serialization qualifies.

Table 2. Decode cost by evidence group. All speedups are baseline/speculative latency (×; higher is faster), with frozen prompt estimators and 95% CIs. Historical agreement does not qualify direct IDs. The last block uses its own fresh qualification and timer; original Figure 3 gates have no matched latency.

Path / workload	Speedup [95% CI]	Endpoint	Pairs	Faster prompts	Class
Historical text-path timing					
M4 representative 24Q	0.842 [0.809, 0.877]	Text-path	72/72	1/24	Primary
M4 HumanEval fixed-$k = 2$	1.387 [1.351, 1.422]	Text-path	90/90	30/30	Primary
CPU to RTX 5070 Ti Laptop	0.404 [0.390, 0.418]	Text-path	90/90	0/30	Secondary
RTX 5070 Ti Laptop to itself	0.650 [0.629, 0.673]	Text-path	90/90	0/30	Secondary
B570 serialized fallback	0.3087 [0.2927, 0.3246]	Text-path	60/90	0/30	Diagnostic
358H direct-ID-qualified complete-loop timing					
358H CPU draft	0.734 [0.693, 0.771]	Direct-ID	90/90	0/30	Primary
358H NPU draft	0.352 [0.333, 0.369]	Direct-ID	90/90	0/30	Primary
New RTX direct-ID-qualified matched instrumented timing					
New RTX serialized latency	0.646 [0.630, 0.660]	Direct-ID	90/90	0/30	Matched cost

Stage	Design or comparator	Complete generated-ID endpoint
Discovery	12 prompts × 3 runs; select target-only UB1 as H1.	UB1 predicts the serialized trajectory in **36/36** runs.
Hypothesis freeze	After discovery; fix H1, reference, tasks and endpoint.	Before reserved reference/predictor inference; H2 remains unrun.
Prediction freeze	After pre-live inference; seal all 72 UB1 arrays.	Before the first reserved live outcome; no prediction edits.
Live test	Compare each of 72 live serialized runs with both frozen paths.	UB1 predictor/live **72/72**; UB128 reference/live **60/72 (BLOCKED)**. Four prompts (12 runs) are reference-divergent.
Divergent case	HumanEval/134; first difference at generated-ID index 88.	Reference ID 914 → predictor/live ID 1946 in all 3 runs.

Figure 2. **B570 prospective prediction.** Hypothesis and full-array freezes are separate stages. The 24 prompts × three repeat-stable runs test complete capped trajectories, including four non-reference predictions. Reported ordering and retained arrays are reviewed in the appendix; prediction success does not restore qualification.

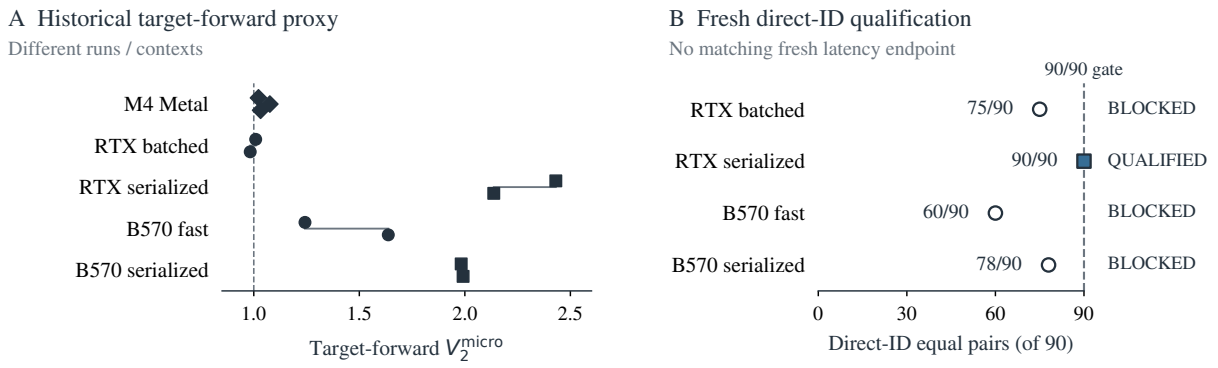
B570 matches 60/90 fast and 78/90 serialized; both remain blocked. Historical B570 text-path 60/90 and fresh fast direct-ID 60/90 are different comparisons. Repeat stability does not imply reference equality. These qualification-only campaigns have no matched latency; Figure 3A/B cannot form a speed/qualification tradeoff. The separate new RTX timing study below does not certify historical timing.

5.2 B570 prediction and sampled live intervention

A target-only harness first reproduces 90/90 UB128 reference sequences. H1 target-only UB1 matches 36/36 sequences on 12 exposed discovery tasks; its frozen tree selects H1, leaving H2 unrun. After conservative exposure exclusion, 24 reserved tasks are selected. H1 freezes before reserved inference and all 72 full predictions freeze before reserved live. Prediction/live agrees 72/72 over three stable runs per prompt, including four reference-divergent prompts (12 runs); the same-set UB128 reference agrees only 60/72 and remains *BLOCKED*. All these reserved arrays reach 128 IDs. The predictor could have failed on any reserved sequence; its success establishes predictive path sufficiency on this B570 stack, not a unique numerical cause. The earlier serialized 78/90 gate has a different task set and reference comparison.

A separately frozen intervention uses 24 new reserved prompts × three. Each sampled checkpoint is the first verification block beginning with at least 64 committed IDs. Before UB1 versus UB128 target verification, the branches share byte-checked KV, prefix, position, fixed proposals and cloned sampler state. Raw scores change at 72/72 forks, but raw top-1 and actual immediate decisions change at 0/72; the 195 paired decision events are correlated within prompts. Subsequent full trajectories differ in 18/72 runs across 6/24 prompts.

The alternate branch restores the fork's draft state and continues with its UB128 target context; later proposals follow that branch's own live state. The secondary continuation



A: path shapes only; all 11 measured cells. B: open = BLOCKED; filled square = QUALIFIED.

Figure 3. **Historical target-forward proxy and fresh direct-ID qualification are separate evidence.** A: historical V_2^{micro} cells from different runs and contexts; shapes encode M4 reference (diamond), fast/batched (circle), and serialized (square) paths, with no qualification encoding. Segments span measured contexts, not uncertainty. The proxy is not complete R_{target} or end-to-end speedup. B: fresh HumanEval30 direct-ID sequence equality, with no matching fresh latency endpoint; only RTX serialization reaches the frozen 90/90 gate. Open circles mean BLOCKED; the filled square means QUALIFIED. The panels are not a matched cost–qualification experiment. Fresh direct-ID qualification does not retrospectively authorize historical timing. Results are path/runtime scoped; contexts and references differ.

differences therefore do not measure an immediate decision effect at the sampled block or show that a one-block switch restores the reference. Under the frozen policy, scores and later trajectories change, but no immediate effect is detected at the sampled states. This does not rule out partition effects elsewhere in generation, identify one responsible operation or explain RTX. Existing-record analysis confirms numerical change, rather than metadata change: across 216 correlated raw-logit rows, median absolute top-1 score change is 0.0695 and maximum 2.3503. Retained top-two logits give minimum margins 0.2291 (UB1) and 0.2003 (UB128); no raw top-1 changes. The six sensitive prompts first diverge 10–50 positions after the fork, with stable capped endpoints. Appendix P gives all positions and clone/R0 controls.

5.3 RTX independent proposal and replay prediction

The separately frozen RTX H1 discovery tests target-only UB2 against native batched live and matches only 24/36, failing its 36/36 gate. No hypothesis is selected, its H2 interface is unavailable, and reserved execution is zero. This failure remains distinct from subsequent V2 work.

V2 implements independent draft-only P and target-only R processes. P constructs proposals using H2's own prior feedback; R performs verification, acceptance/correction/bonus and rollback. A fresh R_ONLY loads only target weights and consumes the sealed tape sequentially, computing its own decisions without future live answers. Original CLI live generates its own draft and imports no tape. Code, model-loading and RPC records support application/process/API independence; all programs share llama.cpp/GGML/CUDA numerical routines.

Discovery H2/live and H2/R_ONLY both pass 36/36. Following F0/F1/F2 freezes, complete predictions and tapes precede reserved live on another 24-task set × three: 288 executions across reference, H2, R_ONLY and live. H2/live, R_ONLY/live and H2/R_ONLY each match 72/72, whereas this study's UB1 reference/live matches 60/72. Four tasks (/17, /33, /14, /22) diverge in all repetitions. Live endpoints contain 63 earlier-EOG runs and nine capped runs; "complete" does not mean all reach 128. The recorded proposals and target path suffice to reconstruct these trajectories; H2 construction still uses drafting. This establishes program-level prospective reproduction. No target-only UB2 competitor was tested on this same V2 reserved set, so it does not establish proposal/history necessity. Appendix M separates allowed proposals from locally recomputed decisions and documents existing rejection/bonus witnesses.

5.4 Matched cost on a newly qualified RTX path

A separate study builds and freezes an instrumented frontend, then obtains fresh serialized/reference direct-ID equality of 90/90 on 30 prompts × three × two conditions (180 timed runs). The prompt estimator is median target time divided by median serialized time, followed by a GM and fixed prompt bootstrap: 0.646× [0.630, 0.660], with 0/30 prompt ratios above one (Table 2). Serialization is slower within this qualified measured boundary.

The timer includes draft and target computation, per-request sampler/generation state, acceptance/correction/bonus, synchronization, state management and related log-

ging/response work through slot release; loading, initial prefill and teardown are excluded. Committed-ID/debug logging is enabled. The original 10,000-sample empty-hook probe has p99 0.6 μs against thresholds of 10 μs and 1% of the shortest R0 interval (49.9572 ms). It bounds clock-call overhead, not all logging cost. In the post-study analysis, the condition medians across prompt-median absolute times are 1907.9 ms target-only and 2923.3 ms serialized (Appendix P); a shared executable does not make condition-dependent logging work cancel. This new result neither replaces historical RTX 0.650 nor supplies a full batched-to-serialized speed tradeoff: no qualified batched timing exists in this same study.

5.5 Stateful realization

On 358H, isolated NPU draft throughput exceeds CPU at tested contexts, but both integrated direct-ID-qualified paths are slower than target-only. Complete committed progress is $C = 2.4631$ and 2.4584, including correction/bonus. Complete target work consumes 87.34% of CPU-path time. The NPU synchronous OpenVINO region combines full-prefix replay/reconstruction and proposal, consuming 58.35%; these costs are inseparable. Residual orchestration is 0.04% and 0.02%.

Figure 4 describes the measured interface, not NPU capability or a measured persistence benefit. Arithmetic totals give CPU $(D_{\text{real}}, R_{\text{target}}, X) = (0.4304, 2.9767, 0.00124)$ and NPU $(4.1251, 2.9435, 0.00130)$, yielding $0.7227\times$ and $0.3477\times$ in Equation 2; these are total-time ratios rather than Table 2's prompt GMs. With measured progress, target-work and baseline totals fixed, nonnegative draft and residual terms imply $S \le C/R_{\text{target}}$: $0.8275\times$ CPU and $0.8352\times$ NPU, even if both other terms vanish. Thus draft-only savings cannot reach parity under this accounting. This post-study arithmetic bound is not a prompt-GM bound or an optimization experiment; changing progress or target execution invalidates its fixed assumptions. The later persistent-state/wider-target candidate stops at qualification without formal performance or energy.

6 From Measured Constraints to Runtime Decisions

Batch-shape sensitivity is already known. B570 adds a prospective test that could fail on any reserved ID. The four correctly predicted non-reference trajectories show that the predictor does more than echo the reference. This path must still be rejected as a lossless replacement for that reference. The fork's unchanged sampled decisions limit the causal explanation, and RTX's failed H1 prevents treating the target-only rule as portable. Its later P/R reproduction uses a different input contract.

The cost results give a second set of decisions. New RTX serialization passes its own direct-ID gate but is slower within its measured boundary. On 358H, removing all draft and residual cost still cannot reach parity at fixed progress and measured target work. An improvement would therefore require changing progress or target work; moving or accelerating the draft alone is insufficient. The M4 study below tests whether continuation control improves a qualified fixed path. These decisions depend on each study's comparator and establish neither a common numerical cause nor a performance law across stacks.

7 Qualification and Realized-Cost Runtime Control

We examine qualification and continuation in separately frozen runtime studies. The 358H candidate stops at direct-ID qualification. On M4, a prospectively direct-ID-qualified path permits a test of a simple cost rule: Guard remains faster than target-only, but fixed speculation is faster in the post-study paired secondary comparison; no incremental control benefit is shown. The studies test the decisions in Section 6; neither establishes a successful optimization intervention.

7.1 An intervention stopped by qualification

The 358H component measurements motivated persistent draft state and a wider target execution setting. The implementation ran the target and draft backends in separate processes. Synchronous draft calls, including communication and state work, remained inside the target worker's post-prefill timer. The implementation then had to pass correctness qualification. CPU persistent-state restore/reconcile correctness could not be established under the tested representation; the Optimum NPU state-capable path was unavailable with the frozen dynamic-shape IR. These are implementation boundaries, not evidence that the devices cannot retain state.

The target setting `n_ubatch=2` (UB2) passed the small gate but failed full qualification. Correctness-only follow-up controls on Qwen3-8B Q8_0, llama.cpp b10839, and Vulkan0 reproduced different greedy tokens under UB1 and UB2 even with identical exact prefix tokens. Request-order and separate-process controls reproduced the target-only divergence. Replaying frozen proposals also reproduced the speculative divergence without live NPU draft inference. The latter depended on earlier target execution history; the controls found no position drift or specific rollback-boundary bug. The responsible floating-point or backend operation remains unidentified.

Under this frozen tested target stack, UB2 did not preserve the strict direct-ID output contract. Only the target refer-

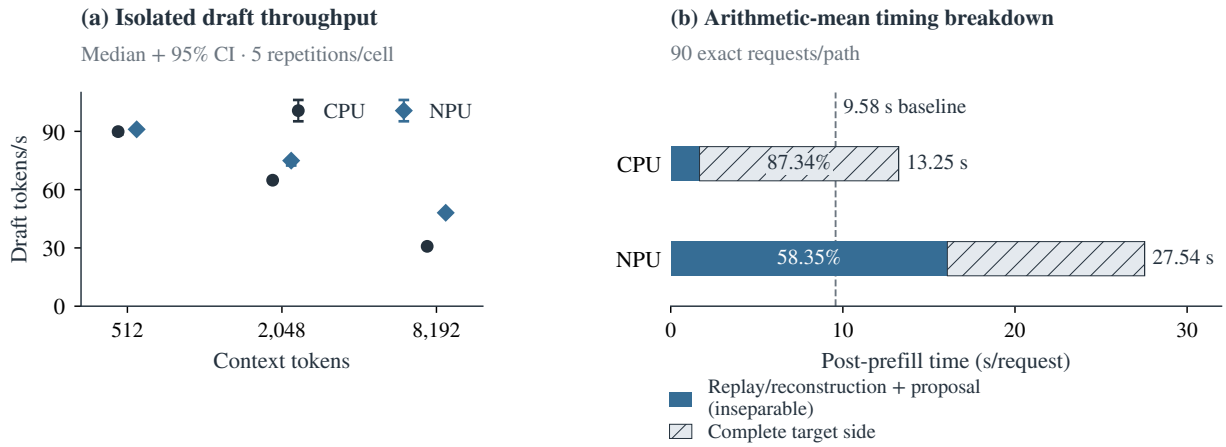


Figure 4. **Isolated draft advantage and complete 358H cost.** A: five-repeat steady-state draft throughput with repetition-bootstrap intervals. B: arithmetic mean post-prefill seconds over 90 exact observations per path, against the shared baseline. NPU's 58.35% region combines full-prefix replay/reconstruction and proposal; it is inseparable. CPU complete target work is 87.34%. These bars differ from the prompt-GM speedups in Table 2; persistence savings are unmeasured.

ence remained in the formal registry, leaving no qualified intervention pair to evaluate. No formal R2 performance run occurred, and the follow-up diagnostics are invalid for performance claims. Qualification excluded an intervention motivated by the measured bottlenecks before its speed was evaluated.

7.2 A prospective cost guard on M4

The M4 experiment asks whether a frozen rule based on already-realized complete cost can control continuation while preserving exact output and profitable execution against target-only. It does not assume that adding control improves on fixed speculation. We retain the Qwen3-8B/0.6B Q8_0 pair on llama.cpp Metal, fixed $k = 2$, greedy generation, and a 128-token cap. Guard begins on the same speculative path as Always-Spec and can fall back permanently to target-only within a request. It changes neither proposal width nor device placement.

At each eligible completed round, Guard divides all elapsed charged generation cost by committed output progress C. This count includes the initial target seed and committed correction or bonus tokens, but excludes rejected proposals. Guard continues if this cumulative cost per token is at most $(1 - m)R_{\text{ref}}$, a target-only reference frozen from calibration; otherwise it falls back. If a decision is reached with zero committed progress, Guard also falls back. Before each control decision, invalid candidate suffix state is removed from the target and draft contexts, preserving committed history for target-only continuation. The enclosing post-prefill clock includes generation-loop initialization, seed-token sampling, drafting, target work, synchronization, state reconciliation, control, and transition costs. Model loading, sampler/speculator setup, initial prompt prefill, and the speculative begin hook precede the timer; the result is not whole-request latency.

The reference is the median across calibration prompts of each prompt's median target latency divided by its median output-token count. The preregistered candidate grid varies the first decision round and margin. Selection minimizes calibration geometric-mean latency relative to the offline minimum of target-only and Always-Spec among exact candidates, with frozen tie-breaks. The selected rule uses margin 0 and first decides after 8 completed speculative rounds; it then decides after each later round while speculation continues. A request ending earlier makes no decision. The online policy receives neither task/family labels nor future output length, current-request target-only latency, or oracle outcomes.

7.3 Held-out evaluation

Historical M4 outcomes informed the study design. Prospective here means that the policy was frozen before the reserved prompt outcomes were measured, not that the design was uninformed by earlier campaigns. IDs marked as previously executed or outcome-exposed in the frozen inventory were excluded before deterministic hash-ranked selection. Disjoint splits contain 40 calibration prompts, 16 qualification prompts, and 60 held-out prompts, with equal

Table 3. M4 ratios of paired prompt medians. Both speculative policies preserve 180/180 direct-ID sequences. Speedup and relative latency have opposite favorable directions. The oracle selects the lower of two measured fixed-mode medians with hindsight. Guard/Always is post-study secondary; values above one mean Guard is slower.

Ratio (×)	GM	95% prompt CI
Speedup: above one favors speculation		
Target / Guard	1.3114	[1.2487, 1.3787]
Target / Always-Spec	1.3469	[1.2867, 1.4088]
Relative latency: lower is better		
Guard / offline oracle	1.0278	[1.0137, 1.0446]
Post-study paired secondary latency ratio		
Guard / Always-Spec	1.0271	[1.0130, 1.0436]

HumanEval and Spec-Bench counts within each split. The policy and reference were frozen before qualification and formal measurement; the appendix documents the audited recovery of a freeze-hash metadata defect. Both Always-Spec and Guard passed 48/48 direct-token comparisons in qualification.

The prospective M4 study runs 3 repetitions of target-only, Always-Spec, and Guard on each held-out prompt, randomizing condition order within prompt and repetition: 540 observations. Both speculative conditions preserve 180/180 complete direct-ID sequences, with zero mismatches. For each prompt and condition, analysis takes the median latency across repetitions, then the geometric mean of prompt latency ratios. This ratio of condition medians differs from the historical M4 text-path hardware estimator. Confidence intervals use 10,000 prompt bootstrap resamples with seed 20260924; repetitions are not independent sampling units.

Guard is profitable against target-only on this held-out workload (Table 3). Fixed speculation has the higher aggregate speedup point estimate. A reviewer-requested post-study paired secondary analysis of the frozen prompt medians gives Guard/Always-Spec latency 1.0271 [1.0130, 1.0436]. The interval is above parity: Guard had higher latency on this comparison. By prompt median, Guard is faster on 24 prompts, Always-Spec on 36, with 0 ties. Guard falls back in 30.6% of its 180 request executions; this measures control activity, not the fraction of beneficial decisions. Guard's geometric-mean latency relative to the offline oracle is 1.0278×. That oracle chooses the lower target-only or Always-Spec median after measurement and was unavailable to the runtime. It is a descriptive hindsight comparator, not an achievable-policy guarantee.

A post-study reconstruction of all original medians finds target-only faster than Always on only 2/60 prompts, both Spec-Bench. Always/oracle is 1.0007×: little opportunity for choosing between these two measured fixed modes. Guard falls back in all six executions of those two prompts but still loses to target-only on their medians. Guard also remains slower in a family-stratified paired bootstrap sensitivity analysis; Appendix P reports this separate new interval. These are hindsight comparisons, not transition-cost estimates.

Guard preserves IDs but supplies no incremental benefit over Always-Spec. Fallback activity does not establish beneficial decisions or risk reduction. No runtime energy claim is made. Policy transfer and stable small-stratum effects remain unestablished; the archived policy-freeze defect and original stopped pipeline markers are documented separately.

8 Related Work

Performance and heterogeneous execution. Speculative decoding amortizes target computation under its algorithmic assumptions (Leviathan et al., 2023). Prior work already documents verification and workload limits: Liu et al. (2026b) study verification dominance and batch effects in vLLM; Kim et al. (2026) account for conditional execution cost and runtime control; Lyu et al. (2026) derive edge–cloud profitability conditions. On consumer Apple silicon, Chordiya (2026) report gains and slowdowns across five backend configurations, diagnose ineffective batch-parallel verification in one quantized Metal path, test output agreement, and observe target-only batch-shape argmax changes. Slowdown and correctness gating have already been studied, as has heterogeneous placement: Dovetail, DuoDecoding, and Mirror-SD study GPU/CPU or GPU/NPU paths (Zhang et al., 2025; Lv et al., 2025; Bhendawade et al., 2025); Ruiz y Mesa et al. (2026) combine an analytical cost model with compiler-oriented CPU/GPU partitioning, and Lever jointly optimizes draft, verification, and CPU/NPU execution for a smartphone target (Wang et al., 2026). State reuse has prior art in compressed draft KV memory (Yuan et al., 2026). Our 358H study measures one interface that combines replay and proposal; it establishes no general placement or persistence result.

Numerical paths and output contracts. Zhang et al. (2026) distinguish ragged-batch synchronization errors from residual numerical mismatches. Yuan et al. (2025) study output variation across system configurations, and LLM-42 addresses deterministic inference through verified speculation (Gond et al., 2026). Numerical path dependence is not new here. We prospectively qualify specific RTX/B570 paths against captured generated IDs, separately from historical text-path timing. For B570, H1 was selected after 36/36 exposed discovery matches, then complete predictions were frozen before 72 reserved live executions; four prompts

match the non-reference prediction in all 12 runs, ruling out a predictor that simply echoes the reference on this set. H2 was unrun in that original B570 tree. Subsequent same-state B570 forks test sampled immediate and continuation effects separately; RTX's H1 fails, while its independently frozen P/R replay study tests program-level prospective reproduction using shared numerical routines. This tests path admissibility and predictive sufficiency, not a new numerical phenomenon, unique kernel cause, or competing reserved predictor.

Adaptive speculation and the study boundary. TurboSpec adjusts speculation for serving goodput (Liu et al., 2025); BanditSpec selects configurations online (Hou et al., 2025); ACL HeteroSpec adapts verification from contextual entropy (Liu et al., 2026a). Adaptive control is established prior work. Our M4 Guard is a qualified runtime control test, not a new policy advantage: it beats target-only, but fixed speculation is faster in the post-study paired secondary comparison. Our measurements test prospective direct-ID qualification, frozen trajectory prediction, sampled intervention effects and realized cost on qualified paths.

9 Limitations

The studies cover a narrow Qwen3 family, greedy capped outputs and independently frozen consumer stacks. Hardware, backend, OS and quantization vary together, so the results cannot establish hardware rankings, population failure rates, functional HumanEval pass@k or production-load behavior. Strict-ID compatibility is conditional on each reference, and historical text-path timing remains distinct. The new RTX timer includes debug logging and lacks a qualified batched comparison or logging-free deployment estimate.

B570 H1 has 24 reserved prompts and four reference-divergent trajectories on one stack. The fork samples one checkpoint per new run; persistent UB128 continuation is not a single-block restoration or an explanation of the old four tasks. RTX H1 fails discovery, and V2 H2 uses drafting and shared numerical routines. Without a same-set target-only UB2 competitor, proposal/history necessity is unknown. No unique operation or portable numerical cause is identified.

Prompt intervals exclude session and recalibration variation. Events/repetitions are correlated; exposure exclusion covers retained records only. Hashes identify bytes and host/server records report order; neither anonymous ordinals nor Git alone authenticates original absolute time or omitted binaries. 358H runtime raw traces and exact old sequence ZIP are unavailable; preserved arrays and processed cost records remain independently checkable. Its timing region combines reconstruction/replay with proposal, so their separate costs are unavailable. The zero-draft bound holds only at fixed measured progress and target work.

M4 Guard/Always is post-study secondary, not the registered primary endpoint. New family-stratified sensitivity and fixed-mode opportunity retain that status. The oracle chooses two measured modes with hindsight, not arbitrary switching policies. The M4 metadata defect retains original error and stopped markers. Runtime energy and 358H persistence gains remain unmeasured. The energy domains differ, and M4's corrected synthetic ceiling was not executed.

10 Conclusion

On the tested B570 stack, frozen draft-free UB1 predictions match 72/72 reserved capped sequences, including 12 runs of four reference-divergent prompts; reference/live agreement is 60/72. The test predicts complete future trajectories rather than only observing path sensitivity, but lossless compatibility remains blocked. Same-state forks examine intervention effects, and independently orchestrated RTX replay tests program reproduction. The cost comparisons lead to specific decisions: qualified RTX serialization is slower within its instrumented boundary; fixed-progress 358H target work prevents parity even at zero draft/residual cost; M4 Guard loses to fixed speculation despite preserving IDs. Each conclusion is limited to its tested reference and execution path. No universal mechanism or successful portable optimization is established.

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